STAT 479: Causal Inference - SP26 Lecture Notes

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01/20/2026

1 Introduction to Causal Effects

Example Case: Estimate the effect of attending a private high school versus a public high school on income at middle age.
Causal Effect: What is the effect of going to private high school v.s. going to public high school.

Y = Income at middle age.
Z = 1 if the individual attended a private high school.
Z = 0 otherwise.

1.0.1 Simple Comparison:

$$\bar{Y}_1 - \bar{Y}_0 = \$10,000 \tag{1}$$

$\bar{Y}_1$ = Average income for private high school graduates.
$\bar{Y}_2$ = Average income for public high school graduates.

However, the difference might not be causal.

Confounding Factors (Other Factors): The difference might be influenced by:

- Level' education of parents.
- University degree.
- Sample characteristics (e.g., Wisconsin graduates in 1990)

1.0.2 Addressing Imbalance in Covariates

When comparing groups ($Z = 1$ vs $Z = 0$), there is often an **imbalance in** X (covariates), such as the education of parents.

1.0.3 Linear Regression Model

To account for these factors, we use a linear regression model:

$$Y = Z\tau + \beta^T X + \epsilon \tag{2}$$

Where:

$$X = \begin{pmatrix} 1 & X_{11} & X_{21} & \dots \\ 1 & X_{12} & X_{22} & \dots \\ \vdots & \vdots & \vdots & \vdots \\ 1 & X_{1n} & X_{2n} & \dots \end{pmatrix}$$

X_{ii} is for the important influence.
For example:

X_1 for father's education
X_2 for mother's education
...

By solving the problem, we can get $\hat{\tau}$, which is the estimated causal effect obtained by solving the regression.

The expectation of ϵ is set to 0.

$$E(\epsilon|Z, X) = 0 \tag{3}$$

Suppose $\hat{\tau}$ is the estimate from the linear regression model, then:

$$\hat{\tau} = \frac{\sum_{i:z_i=1} \theta_i Y_i}{\sum_{i:z_i=1} \theta_i} - \frac{\sum_{i:z_i=0} \theta_i Y_i}{\sum_{i:z_i=0} \theta_i} \tag{4}$$

Note: $\theta_i = 1$ for all i, we come back to $\bar{y_1} - \bar{y_0}$

By some specific θ_i , we determine who get higher/lower/balanced weight.

2 The Frisch-Waugh-Lovell (FWL) Theorem

Consider the linear regression of Y on (Z, X).The coefficient on Z is numerically equal to the coefficient from the regression of Y on $\tilde{Z}$, where:

$$\tilde{Z} = Z - \mathbf{E}[Z|X] \tag{5}$$

It is also numerically equal to the coefficient from the regression of $\tilde{Y}$ on $\tilde{Z}$, where:

$$\tilde{Y} = Y - \mathbf{E}[Y|X] \tag{6}$$

$$\tilde{Z} = Z - \mathbf{E}[Z|X] \tag{7}$$

In conclusion, $\tilde{Z}$ is the residual from the regression of Z on X. The purpose is to make things easier.

Then we can make the transformation:

$$Y = \tilde{Z}\alpha + u, then : \alpha = \tau \tag{8}$$

3 Regression Coefficients as Weighted Averages

The linear regression estimator of the treatment effect, $\hat{\tau}$, can be written as the **weighted difference** in outcomes between the treated and untreated groups. This means that regression does not directly compare simple averages, but instead the **weighted averages of outcomes**.

using FWL, $\hat{\tau}$ can be written as:

$$\hat{\tau} = \frac{\sum_{i=1}^{n} \tilde{Z}_i Y_i}{\sum_{i=1}^{n} \tilde{Z}_i^2} \tag{9}$$

The weights are defined as:

$$w_i = \frac{\tilde{Z}_i}{\sum_{j=1}^{n} \tilde{Z}_j^2} \tag{10}$$

Then the weights can be substituted into the estimator,

$$\hat{\tau} = \sum_{i=1}^{n} w_i Y_i = \sum_{i:Z_i=1} w_i Y_i + \sum_{i:Z_i=0} (-w_i) Y_i \tag{11}$$

Weighted Difference Representation The expression for $\hat{\tau}$ can be further understood by separating treated and control units. Since $Z_i \in \{0, 1\}$, observations naturally partition into treated ($Z_i = 1$) and untreated ($Z_i = 0$) groups. Writing the estimator in this way highlights that $\hat{\tau}$ compares outcomes across these two groups using regression-induced weights.

In particular, the estimator can be written as

$$\hat{\tau} = \sum_{i:Z_i=1} w_i Y_i - \sum_{i:Z_i=0} (-w_i) Y_i,$$

where treated units receive positive weights and control units receive negative weights.

These weights depend on the residualized treatment $\tilde{Z}_i$, meaning they reflect how unexpected treatment assignment is given the covariates. As a result, regression does not compare simple averages, but instead forms a weighted difference between treated and control outcomes.

3.0.1 Balancing Between Treated and Control Groups

A key implication of the weighted representation of $\hat{\tau}$ is that linear regression balances the treated and untreated groups with respect to the included covariates X. Importantly, this balance is achieved through the regression weights rather than by directly matching or removing observations.

The residualized treatment is defined as

$$\tilde{Z}_i = Z_i - E[Z \mid X_i],$$

which represents the part of treatment assignment that cannot be explained by the covariates. Because $\tilde{Z}_i$ is a residual from regressing Z on X, it has no systematic relationship with X. As a result, positive and negative values of $\tilde{Z}_i$ cancel out when multiplied by X, implying

$$\sum_{i=1}^{n} \tilde{Z}_i X_i = 0.$$

Since the regression weights are proportional to $\tilde{Z}_i$,

$$w_i = \frac{\tilde{Z}_i}{\sum_{j=1}^{n} \tilde{Z}_j^2},$$

the same cancellation occurs when covariates are weighted by w_i. This leads to the global balance condition

$$\sum_{i=1}^{n} w_i X_i = 0,$$

which means that, after weighting, covariates do not differ on average across the sample.

To see how this creates balance between treated and control groups, we separate the weighted sum by treatment status:

$$\sum_{i:Z_i=1} w_i X_i + \sum_{i:Z_i=0} w_i X_i = 0.$$

Rearranging gives

$$\sum_{i:Z_i=1} w_i X_i = \sum_{i:Z_i=0} (-w_i) X_i.$$

This equality shows that the weighted average of covariates among treated units is exactly the same as the weighted average among control units (after flipping the sign of control weights). In other words, regression assigns weights so that the treated and control groups are made comparable in terms of X.

Interpretation Linear regression adjustment can be understood as a balancing procedure. Rather than comparing raw outcome averages, OLS reweights observations so that treated and control groups look the same with respect to the included covariates. The estimated treatment effect $\hat{\tau}$ is then computed as the difference in weighted average outcomes between these balanced groups.

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Thursday, Jan 29th 2026

1 Setup – Potential Outcomes Framework

Consider a finite population of N units ($i = 1, \ldots, N$).

For each unit i we define two **potential outcomes**:

- $Y_i(1)$ = outcome if unit i receives treatment
- $Y_i(0)$ = outcome if unit i receives control

These are **fixed** (non-random) but **unknown**.

We observe only one outcome per unit:

$$Y_i^{obs} = Z_i Y_i(1) + (1 - Z_i) Y_i(0)$$

where the binary indicator $Z_i = 1$ if treated, $Z_i = 0$ if control.

Fundamental problem of causal inference: we never observe both $Y_i(1)$ and $Y_i(0)$ for the same unit. The *factual* outcome is observed and the *counterfactual* outcome is unobserved.

2 Target Causal Quantity – PATE

The quantity we want to estimate is the finite **Population Average Treatment Effect** (PATE):

$$\tau = PATE = \frac{1}{N} \sum_{i=1}^{N} (Y_i(1) - Y_i(0)) = \frac{1}{N} \sum_{i=1}^{N} \tau_i$$

where $\tau_i = Y_i(1) - Y_i(0)$ is the individual causal effect of treatment Z on individual i. This individual treatment effect is not estimable because we never observe both potential outcomes for the same individual (which is the fundamental problem of causal inference). What is estimable (under

assumptions like random assignment) is an average causal effect, such as the PATE. This is a **fixed number** (not a random variable) — it describes the average effect across the entire finite population in the study.

3 Neymanian vs Fisherian Inference Approaches

	Neymanian	**Fisherian**
Causal estimand	$\tau_i = Y_i(1) - Y_i(0)$	$\tau_i = \tau$ for all i
Target parameter	$\tau = \frac{1}{N}\sum(Y_i(1) - Y_i(0))$	τ (constant)
Population	Finite population ($N < \infty$)	Superpopulation / infinite
Assumption on effects	Heterogeneous allowed	Homogeneous effect assumed
Confidence interval justification	Randomization distribution	Usually model-based
Works when	$N \to \infty$	Any N (but strong assumptions)
Typical notation	$[\hat{\tau} \pm 1.96\sqrt{\widehat{Var}}]$	Similar but different variance

4 Estimator under Completely Randomized Design

Under completely randomized assignment with n_1 treated and $n_0 = N - n_1$ control units, the natural estimator is the difference-in-means:

$$\hat{\tau} = \bar{Y}_T - \bar{Y}_C = \frac{1}{n_1}\sum_{i:Z_i=1} Y_i^{obs} - \frac{1}{n_0}\sum_{i:Z_i=0} Y_i^{obs}$$

4.1 Unbiasedness (Neyman–Randomization Perspective)

$$E[\bar{Y}_T] = \frac{1}{N}\sum_{i=1}^{N} Y_i(1) = \bar{Y}(1)$$

$$E[\bar{Y}_C] = \frac{1}{N}\sum_{i=1}^{N} Y_i(0) = \bar{Y}(0)$$

$$\Rightarrow \quad E[\hat{\tau}] = \bar{Y}(1) - \bar{Y}(0) = \tau$$

→ The difference-in-means estimator is **unbiased** for the PATE under randomization — **no modeling assumptions** needed.

5 Neyman's Exact Variance under Randomization

The exact variance of $\hat{\tau}$ under the randomization distribution is:

$$Var(\hat{\tau}) = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{S_\tau^2}{N}$$

where
$\mathrm{S}_1^2 = \frac{1}{N-1}\sum_{i=1}^{N}(Y_i(1) - \bar{Y}(1))^2$
$S_0^2 = \frac{1}{N-1}\sum_{i=1}^{N}(Y_i(0) - \bar{Y}(0))^2$
$S_\tau^2 = \frac{1}{N-1}\sum_{i=1}^{N}((Y_i(1) - Y_i(0)) - \tau)^2$
The last term is **unobservable** (we never see both potential outcomes).

6 Conservative Variance Estimator (Most Common in Practice)

Since $S_\tau^2 \geq 0$, we have the upper bound:

$$Var(\hat{\tau}) \leq \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0}$$

We replace the population variances with sample variances:

$$s_1^2 = \frac{1}{n_1 - 1}\sum_{i:Z_i=1}(Y_i^{obs} - \bar{Y}_T)^2, \quad s_0^2 = \frac{1}{n_0 - 1}\sum_{i:Z_i=0}(Y_i^{obs} - \bar{Y}_C)^2$$

Conservative (upper-bound) variance estimator:

$$\widehat{Var}(\hat{\tau}) = \frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}$$

7 Large-sample Confidence Interval (Neyman + CLT)

Under mild conditions (as $N \to \infty$, $n_1/N \to p$, $n_0/N \to 1-p$, $p \in (0,1)$), the central limit theorem for the randomization distribution gives:

$$\frac{\hat{\tau} - \tau}{\sqrt{Var(\hat{\tau})}} \; d \; \mathcal{N}(0,1)$$

Approximate 95% confidence interval using the conservative variance estimator:

$$\hat{\tau} \ \pm \ 1.96\sqrt{s_1^2 n_1 + s_0^2 n_0}$$

Important: This normality comes from the **randomization distribution** (not from assuming outcomes are normally distributed).

8 Comparison: Neyman vs. Classical t-test

- **Neyman/Randomization-based**: variance justified by design (random assignment); conservative estimator common; finite population target
- **Classical two-sample t-test**: assumes two random samples from superpopulations; normality or large-sample assumption on outcomes themselves

Neymanian inference is generally considered more appropriate for randomized experiments because it relies only on the design, not on modeling assumptions about the outcome distribution.

Key slogan: "Let the randomization do the work."

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Grace Kachelski

2/3/26

1 Neymanian Inference in an A/B Test: Notification Timing

- This is covered in the Python code, which was translated from the R code given in class.

2 Fisherian Inference

A null hypothesis is called "sharp" if, under H_0, our observed data y^{obs} can be used to calculate both $Y_i(1)$ and $Y_i(0)$.

- Example of a sharp null hypothesis: H_0: $Y_i(1) = Y_i(0)$ for all i.

- Example 2: H_0: $Y_i(1) = 2Y_i(0)$ for all i.

For $Z_i = 1$: $Y_i(1) = Y_i^{obs}$ and $Y_i(0) = \frac{1}{2}Y_i^{obs}$

For $Z_i = 0$: $Y_i(1) = 2Y_i^{obs}$ and $Y_i(0) = Y_i^{obs}$

- This null hypothesis is sharp because not only is $Y_i(1)$ equal to $Y_i(0)$, but these are also both equal to Y_i^{obs}.

Example of a null hypothesis that is NOT sharp: H_0: $\tau = \frac{1}{N}\sum_{i=1}^{N}\tau_i = 0$

- This null hypothesis is not sharp because it doesn't specifically tell us about $Y_i(1)$ and $Y_i(0)$; could be a range of values.

- Instead, this type of null is called a "weak null".

i	$Y_i(0)$	$Y_i(1)$	τ_i
1	0	1	1
2	0	1	1
3	0	0	0
4	0	-1	-1
5	0	-1	-1

Table 1: Full Data

z_i	Y_i^{obs}
1	1
1	1
1	0
0	0
0	0

Table 2: Observed Data

- Example 2:

We are unable to recover the full data from the observed data here: our null is weak.

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By Natalie Gjerseth and Amooyla Mammula

Fisherian Inference (continued)

It uses likelihood, maximum likelihood estimation, and p-values to measure evidence against a null hypothesis.

Example:

- 6 classrooms, 3 are assigned to the new way of teaching, 3 assigned to the old way of teaching.
- The classrooms were randomly selected.
- Probability of any 3 of 6 classrooms being in the new way of teaching group is
    - (6 choose 3)

Question: Does the new way of teaching increase exam scores?

Classroom	New Way of Teaching?	Z	Avg Exam Score
1	Y	1	97
2	Y	1	96
3	N	0	96
4	N	0	91
5	Y	1	95
6	N	0	96

- (We cannot do Newmarian Inference here, due to small sample size.)

Fisherian Inference:

- We make a Science Table, also known as ‘S tables’
    -

	$Y_i(0)$	$Y_i(1)$
1	?	97
2	?	96
3	96	?

4	91	?
5	?	95
6	96	?

- Test Sharp Null Hypothesis:
  - H_0: $Y_i(0) = Y_i(1)$ for all i
  - H_1: $Y_i(1) >= Y_i(0)$ for all i and the inequality is strict for at least one i
    - H_1 is saying 'New way is at least as good as old way'
- Steps:
  1. Fix alpha, type 1 error: 5%, 10%, etc
  2. Choose test statistic
  3. Decide on hypothesis
  4. Calculate p-value (how much evidence we have against H_0 , the lower the p-value the more evidence we have to reject H_0)
- P-value: the probability under the null hypothesis that the test statistic is equal to or more extreme than the observed value in the direction of alternatives.
  - Example: H_0: $\mu = 0$, H_1: $\mu > 0$
  - Sample average = 1.1, standard error = 2, sample size = 40
  - Under t-test, (avg-0)/SE = 1.1 / 2 = 0.55
  - H_0: Pr(z >= 0.55)
- Test statistic for classroom example
  - T(Z,S) = avg under group 1 - avg under group 0
    - = (97+96+95)/3 - (96+91+96)/3 = 1.66
- Question: We want to calculate the probability under H_0 that T(Z,S) is >= 1.66
  - We calculate the randomization distribution of T(Z,S) induced by the random assignment of Z.
  - 6 choose 3 values of z, 20 possible values
  - P-value: ∑ II (T(z,S)=1.66) x 1/(6 choose 3)
- Table of some possible values:
  -

	z	$Y_i(0)$	$Y_i(1)$	z	$Y_i(0)$	$Y_i(1)$
1	1	?	97	1	?	97
2	1	?	96	1	?	96
3	1	?	96	0	96	?
4	0	91	?	1	?	91
5	0	95	?	0	95	?
6	0	96	?	0	96	?

- Fisherian only works for small samples. If we have a bigger sample, let's say 60 classes, then we need to change to (60 choose 30) - which we can't really do

- Question: What is the smallest possible value of any p-value in this case? (6 choose 3)
    - Answer: 1/20 = 0.05, only 20 values

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2/10/2026

1 Fisherian Randomization Inference (continued)

1.1 Setup

Consider an example with 8 people. We want to assign exactly 4 people to the treatment group and the remaining 4 to the control group.

Based on the observed data and the random assignment defined in the in-class R code, we have

i	$Y_i(0)$	$Y_i(1)$
1	?	30
2	25	?
3	24	?
4	?	25
5	22	?
6	?	28
7	?	26
8	20	?

1.2 Sharp Null Hypothesis: No Effect

Suppose our sharp null hypothesis is

$$H_0 : Y_i(1) = Y_i(0) \text{ for all } i = 1, 2, \ldots, 8.$$

1.3 Test Statistic

We use the test statistic:

$$T(Z, S) = (\text{avg of } Y \text{ in group } Z = 1) - (\text{avg of } Y \text{ in group } Z = 0).$$

From the observed assignment, we compute

$$T_{obs} = 4.5.$$

1.4 Greater-Than Alternative

If the alternative hypothesis is

$$H_1 : Y_i(1) > Y_i(0) \text{ for some } i,$$

then the p-value is

$$\Pr\big(T(Z,S) \geq 4.5\big).$$

1.5 Two-Sided Alternative

If instead

$$H_1 : Y_i(1) \neq Y_i(0) \text{ for some } i,$$

then the p-value is

$$\Pr\big(|T(Z,S)| \geq 4.5\big).$$

More generally, for a two-sided test,

$$p = \Pr\big(|T_{stat}| \geq |T_{obs}|\big).$$

Note: Multiplying the one-sided p-value by 2 is not necessarily correct, since the randomization distribution is not guaranteed to be symmetric.

1.6 Science under H_0

Because the null hypothesis is *sharp*, our observed data Y_{obs} can be used to calculate both $Y_i(1)$ and $Y_i(0)$. Under H_0, the science is fully known:

i	$Y_i(0)$	$Y_i(1)$
1	30	30
2	25	25
3	24	24
4	25	25
5	22	22
6	28	28
7	26	26
8	20	20

1.7 Randomization Distribution

To calculate the p-value, we enumerate all $\binom{8}{4} = \frac{8!}{4!\,4!} = 70$ possible treatment assignments.

For each possible assignment, we compute the test statistic $T(z, S)$. This generates the distribution of our test statistic under H_0.

$$p\text{-value for greater-than alternative} = \frac{\#\{T(z,S) \geq T_{\text{obs}}\}}{70}.$$

The resulting p-value is approximately $0.0286 < 0.05$. Therefore, at 5% significance level, we reject H_0. Because the treatment group is randomly assigned, this provides evidence of a positive causal effect.

Remark: Randomized experiments are considered the gold standard of causal inference because randomization inference provides causal conclusions without modeling assumptions, relying only on the known assignment mechanism.

1.8 Testing Another Sharp Null Hypothesis

Now consider a different sharp null hypothesis:

$$H_0 : Y_i(1) = Y_i(0) + 1 \text{ for all } i.$$

Under the new H_0, the science is

i	$Y_i(0)$	$Y_i(1)$
1	29	30
2	25	26
3	24	25
4	24	25
5	22	23
6	27	28
7	25	26
8	20	21

Again, we enumerate all 70 possible treatment assignments and compute the corresponding test statistic for each to obtain the randomization distribution.

This time, the p-value for greater-than alternative is approximately $0.0714 > 0.05$.

Therefore, we fail to reject this H_0 at the 5% significance level, and 1 is a possible constant additive effect.

Key Insight: The randomization distribution depends on (i) the specific sharp null hypothesis being tested, and (ii) the chosen test statistic.

Sharp Null Hypothesis

$$H_0 : Y_i(1) = Y_i(0) \quad for\ all\ i$$

Equivalently, under a constant additive effect formulation,

$$H_0(\tau) : Y_i(1) = Y_i(0) + \tau \quad foralli.$$

We collect all values of τ for which the null hypothesis is not rejected at significance level $\alpha = 0.05$. The resulting set forms a 95% confidence interval for τ.

This confidence interval is finite-sample exact and randomization-based, obtained by inverting randomization tests under the constant additive treatment effect assumption.

Randomization-Based Confidence Interval

The confidence interval is **not based on a distributional approximation**. It is based on **inverting randomization tests**.

$$\mathcal{C}_{0.95} = \{\tau : p(\tau) \geq 0.05\}$$

To find the lower limit (one-sided 95% CI):

$$p(\tau_L) = 0.05$$

If you had a two-sided test, you would close off both sides.

Why Randomization Matters

If you do not have random assignment, you must discuss bias.

Randomization is the **gold standard** because:

- Identification rests on known, transparent mechanisms.
- Inference is valid in finite samples.
- No untestable assumptions about outcomes are required.
- Sensitivity to hidden bias is zero by design.

Complete Randomization

Suppose:

- N total units
- n are randomly selected for treatment

Consider N people:

$$1, 2, 3, \ldots, N$$

Let $X_1, \ldots, X_N$ denote pre-treatment covariates.

$$\Pr(Z_1 = z_1, \ldots, Z_N = z_N \mid X_1, \ldots, X_N)$$

Probability Distribution over selecting people to go to group 1 v 0 based on only (X$_1, \ldots, X_N$)

Most importantly:

$$(Z_1, \ldots, Z_N) \perp\!\!\!\perp \{Y_i(1), Y_i(0)\}_{i=1}^N$$

That is, assignment is independent of all potential outcomes.

Type I and Type II Error

- Type I error: Reject H_0 when it is true (probability α).
- Type II error: Fail to reject H_0 when it is false.
- Power = 1− Type II error.

Important Conceptual Takeaways

- Randomization inference works in finite samples.
- It does not rely on normal or t-distributions.
- Confidence intervals come from inversion, not asymptotics.
- Without random assignment, bias must be addressed explicitly.

STAT 479: Causal Inference – Student Lecture Notes

Yi Yang

Lecture Date: February 12, 2026

Covariate Adjustment by Linear Regression in a Completely Randomized Experiment

In a completely randomized experiment, causal identification is secured by the assignment mechanism itself. Once the experimenter fixes the number of treated and control units and randomly assigns treatment, the treated and control groups are comparable in expectation along every pre-treatment dimension. This ensures that the difference-in-means estimator is unbiased for the finite-population average treatment effect.

Difference-in-means estimator: $\hat{\tau}_DM = \bar{Y}_1 - \bar{Y}_0$

No regression model or distributional assumption is required for causal identification. The experimental design itself provides identification. However, researchers often run regressions of the form: $Y_i = \alpha + \tau Z_i + \beta^T X_i + \varepsilon_i$ and interpret τ as the treatment effect. This is sensible because regression can improve precision while preserving the causal interpretation under randomization.

Design-Based Framework: Potential outcomes $Y_i(1)$ and $Y_i(0)$ are fixed unknown values. Covariates X_i are fixed characteristics. The only random component is the assignment vector Z. Observed outcomes follow: $Y_obs = ZY(1) + (1 - Z)Y(0)$

Why Adjust for Covariates? Although randomization balances covariates on average, any specific assignment can have chance imbalance. If covariates predict outcomes, this imbalance increases variance. Regression removes predictable variation and produces a cleaner comparison between treatment and control outcomes.

Frisch–Waugh–Lovell (FWL) Interpretation: The regression coefficient on Z equals the coefficient from regressing residualized outcomes on residualized treatment. This removes linear relationships with covariates before comparing treatment and control outcomes.

Unbiasedness: The regression-adjusted estimator remains unbiased under complete randomization, even if the regression model is misspecified, because residualized treatment remains orthogonal to covariates in expectation.

Variance Reduction: If covariates explain outcome variation, regression removes predictable noise and reduces estimator variance. Asymptotically, regression-adjusted variance is less than or equal to difference-in-means variance.

Interpretation: Regression adjustment refines an already valid experimental design. It does not create identification, but improves efficiency by removing noise. The treatment coefficient retains its design-based causal meaning.
Summary: Regression adjustment is best understood as post-randomization blocking. It removes chance covariate imbalance, reduces variance when covariates are predictive, and preserves causal interpretation.

R Code Example: Randomization-Based Inference

```
rand_assign <- function(N, n1, reps = 2000) {
 replicate(reps, {
 z <- rep(0, N)
 z[sample(1:N, n1)] <- 1
 z
 })
}

T_diffmeans <- function(Z, Y) {
 mean(Y[Z == 1]) - mean(Y[Z == 0])
}

rand_test <- function(Z_obs, Y_obs, Tfun, Z_mat) {
 T_obs <- Tfun(Z_obs, Y_obs)
 T_null <- apply(Z_mat, 2, Tfun, Y = Y_obs)
 mean(abs(T_null) >= abs(T_obs))
}
```

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2/17/2026

1 Non-randomized (Observational) Studies

In reality, due to ethical concerns or logistical restrictions, randomization is sometimes impossible. Scientists use observational studies to approximate the balance that randomization would have created.

1.1 Natural experiment

A natural experiment refers to a situation in which researchers do not intervene, but there exists quasi-randomization in treatment assignment. In Card's study (1993), proximity to colleges provides such quasi-randomization in college attendance. In contrast, earlier studies in health science lacked quasi-randomization in the use of hormone therapy, leading to biased estimates.

1.2 Randomized V.S. Non-randomized

In randomized experiments:

$E[X|Z=0] = E[X|Z=1]$ and $P[X=x|Z=0] = P[X=x|Z=1]$

which hold for both observed and unobserved covariates X. In non-randomized settings, even for observed covariates:

$E[X|Z=0] \neq E[X|Z=1]$ and $P[X=x|Z=0] \neq P[X=x|Z=1]$.

1.3 Matching

To approximate the balance:

$E[X|Z=0] \approx E[X|Z=1]$

$P[X=x|Z=0] \approx P[X=x|Z=1]$

We introduce a technique called **matching**. The fundamental idea of matching is as follows: for all i with $Z_i = 0$, find $j(i)$ with $Z_{j(i)} = 1$, so x_i is paired with some $x_{j(i)}$ that is as similar to x_i as possible.

Matching methods differ in how they define similarity between x_i and $x_{j(i)}$. For example, in optimal pair match design, we aim to minimize:

$$\sum_i (x_i - x_{j(i)})^2$$

For vector-valued covariates, minimization task can be based on Euclidean distance:

$$\sum_i ||\vec{x_i} - \vec{x_{j(i)}}||^2$$

In practice, Mahalanobis distance is commonly used:

$$\sum_i (\vec{x_i} - \vec{x_{j(i)}})^T S^{-1} (\vec{x_i} - \vec{x_{j(i)}})$$

where S is the sample covariance matrix of X_is.

Define $M := \{(i, j(i)); Z_i = 1\}$ as the set of paired units. The control group and treatment group have the same number of units. Therefore:

$$\bar{X_T} = \frac{1}{n_M} \sum_{Z_i=1, i \in M} X_i$$

$$\bar{X_C} = \frac{1}{n_M} \sum_{Z_i=1, i \in M} X_{j(i)}$$

1.4 SMD

The standardized mean difference shows how different the means are and allows us to assess balance across covariates.

$$SMD(X) = \frac{\bar{X_T} - \bar{X_C}}{s_X}$$

$$s_X = \sqrt{\frac{s_T^2 + s_C^2}{2}}$$

Note that $SMD = 0$ indicates perfect mean balance and $|SMD| < 0.1$ is regarded as excellent balance.

STAT 479: Causal Inference - SP26 Lecture Notes

[ishaan chauhan]

[02/19/2026]

1 Demonstration of Matching in Observational Studies

Today's lecture focused on the matching problem in observational studies: how to pair treated and control units in a principled way so that outcome comparisons are credible.

2 The Matching Problem

2.1 Setup and Inputs

We have two groups of units:

- Treated units $\{i : Z_i = 1\}$
- Control units $\{j : Z_j = 0\}$

The objective is to find a match $j(i)$ for each treated unit i so that we minimize the total distance:

$$\sum_i d\big(i,\, j(i)\big)$$

where d is an appropriate distance structure between units.

2.2 Feasibility Constraints

For a valid matching, we require:

- $\#\{i : Z_i = 1\}$ is small (number of treated units)
- $\#\{j : Z_j = 0\}$ controls the size of the control pool

Each control unit can only be used once (in 1:1 matching without replacement).

3 Distance Structure

Suppose each unit has covariates $\mathbf{x} = (x^{(1)}, x^{(2)})$, e.g. age, education, degree, marital status.

A natural distance between units i and j is the Mahalanobis distance, which can be decomposed by covariate group:

$$\begin{aligned} d(i,j) = \omega^{(1)} \left(\mathbf{x}_i^{(1)} - \mathbf{x}_j^{(1)}\right)^T \Sigma^{(1)^{-1}} \left(\mathbf{x}_i^{(1)} - \mathbf{x}_j^{(1)}\right) \\ + \omega^{(2)} \left(\mathbf{x}_i^{(2)} - \mathbf{x}_j^{(2)}\right)^T \Sigma^{(2)^{-1}} \left(\mathbf{x}_i^{(2)} - \mathbf{x}_j^{(2)}\right) \end{aligned}$$

where $\omega^{(1)}, \omega^{(2)}$ are weights assigned to each covariate group and $\Sigma^{(k)}$ is the covariance matrix for that group.

3.1 Design Choices

Several design choices arise in practice:

- **1:1 vs. 1:K matching** — match each treated unit to one or multiple controls
- **Different distance structures** — Euclidean, Mahalanobis, propensity score distance
- **Exact matching on a category** — require $\text{Cat}_i = \text{Cat}_{j(i)}$ for all i, so treated and matched control share exactly the same category level
- **Fine balance on a category** — we may not require exact within-pair match, but instead enforce that the frequency of each category in the treated group $\{i\}$ equals the frequency in the matched control group $\{j(i)\}$:

$$\text{freq of Cat in } \{i\} = \text{freq of Cat in } \{j(i)\}$$

4 Evaluation: Standardized Mean Difference (SMD)

After constructing a matched sample, we need to evaluate how well balance was achieved. The primary tool is the **Standardized Mean Difference (SMD)**.

For a covariate X, the SMD is defined as:

$$\text{SMD}(X) = \frac{\bar{X}_1 - \bar{X}_0}{s_X}$$

where $\bar{X}_1$ and $\bar{X}_0$ are the treated and control means, and s_X is a pooled standard deviation. This is computed *before* matching to assess initial imbalance.

After matching, the SMD is recomputed over the matched sample:

$$\text{SMD}_M(X) = \frac{\bar{X}_{1,M} - \bar{X}_{0,M}}{s_X}$$

A successful match substantially reduces the SMD. Common thresholds: $|\text{SMD}| < 0.1$ is considered excellent balance; $|\text{SMD}| > 0.25$ is typically considered unacceptable.

4.1 Why SMD Over Hypothesis Tests?

The SMD is preferred over p-values from t-tests for balance checking because p-values depend on sample size. In large samples, even trivially small imbalances appear statistically significant. The SMD measures *substantive* imbalance regardless of sample size.

5 Key Takeaways

- Matching is a *design* tool: it constructs a comparable dataset before any outcome analysis.
- The objective is to minimize total distance $\sum_i d(i, j(i))$ subject to feasibility constraints.
- Design choices (1:1 vs. 1:K, exact vs. fine balance, distance metric) all affect the quality of the resulting matched sample.
- Covariate balance is evaluated using the SMD before and after matching.
- Good covariate balance implies small bias in matched treatment effect estimators.

STAT 479: Causal Inference - SP26 Lecture Notes

Zhuoming Liu

Feb 26th, 2026

1 Review: Context from Previous Lesson

Before diving into today's derivation, we recall the fundamental distinction between the latent world and observed world.

- **Latent World:** Characterized by variables $(X_i, U_i, Y_i(1), Y_i(0))$.
- **Observed World:** Characterized by variables (X_i, Z_i, Y_i).
- **Target Estimand :** $ATE = E[Y_i(1) - Y_i(0)]$
- **Identification:** An identification result is defined as an equality established between the **target estimand** and the **observed data**.
- **Core Assumptions for Identification:**
  1. SUTVA (Stable Unit Treatment Value Assumption)
  2. Strong Ignorability

2 The Naive Estimand and the Composition of Bias

The "Naive Estimand" represents the raw difference in means from observed data, which often contains bias.

2.1 Derivation from Observed Data

The Naive Estimand is:

$$E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0]$$

By applying **SUTVA**, this can be written in terms of potential outcomes:

$$= E[Y_i(1)|Z_i = 1] - E[Y_i(0)|Z_i = 0]$$

By adding and subtracting the term $E[Y_i(0)|Z_i = 1]$, we can decompose the naive estimand:

$$= \underbrace{E[Y_i(1) - Y_i(0)|Z_i = 1]}_{\text{ATT}} + \underbrace{E[Y_i(0)|Z_i = 1] - E[Y_i(0)|Z_i = 0]}_{\text{bias}}$$

2.2 Conditions for Bias Elimination

- If $Z_i \perp Y_i(0)$, then **bias does not exist**.
  In this case:
  $$E[Y_i(0)|Z_i = 1] = E[Y_i(0)]$$
  $$E[Y_i(0)|Z_i = 0] = E[Y_i(0)].$$
  Therefore,
  $$E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0] = ATT.$$
- **Key Note:** $Z_i \perp Y_i(0)$ holds if the treatment Z_i is **completely random**.

3 Definitions of Causal Estimands

Using the example where Z is smoking and Y is blood pressure:

- **ATE (Average Treatment Effect):** $E[Y_i(1) - Y_i(0)]$
- **ATT (Average Treatment Effect on the Treated):** $E[Y_i(1) - Y_i(0)|Z_i = 1]$
  - *Interpretation:* The average effect for people who smoke.
- **ATC (Average Treatment Effect on the Control):** $E[Y_i(1) - Y_i(0)|Z_i = 0]$

4 Identification via Conditioning on Covariates (X_i)

When treatment is not random, we condition on covariates to achieve ignorability.

4.1 Conditional Independence

By the **ignorability** assumption:
$$Z_i \perp (Y_i(1), Y_i(0))|X_i$$
This independence holds for all X_i.

4.2 Step-by-Step Conditional Identification

We begin with the observed expectation for the treated group:
$$E[Y_i \mid X_i, Z_i = 1]$$
by SUTVA:
$$= E[Y_i(1) \mid X_i, Z_i = 1]$$
by ignorability:
$$= E[Y_i(1) \mid X_i]$$
Similarly, for the control group:
$$E[Y_i \mid X_i, Z_i = 0] = E[Y_i(0) \mid X_i]$$

4.3 Recovering ATE

Taking the expectation over the distribution of X_i allows us to recover the population ATE:

$$\begin{aligned} E[E[Y_i|X_i, Z_i = 1] - E[Y_i|X_i, Z_i = 0]] &= E[E[Y_i(1) - Y_i(0)|X_i]] \\ &= E[Y_i(1) - Y_i(0)] \\ &= ATE \end{aligned}$$

4.4 Interpretation

Using the example where X is **gender**, Z is **smoking**, and Y is **blood pressure**:

- $E[Y_i|X_i, Z_i = 1] - E[Y_i|X_i, Z_i = 0]$: Represents the differences between smokers and non-smokers for the **same gender**.
- The outside expectation $E[\dots]$ is taken over X_is, which accounts for **both genders** (expectation over the population distribution of X).

4.5 Identifying ATT and ATC via X_i

- $ATT = E[E(Y_i|X_i, Z_i = 1) - E(Y_i|X_i, Z_i = 0)|Z_i = 1]$
- $ATC = E[E(Y_i|X_i, Z_i = 1) - E(Y_i|X_i, Z_i = 0)|Z_i = 0]$

5 Propensity Score (e_i)

The propensity score reduces the dimensionality of X_i.

5.1 Definition

$$e_i = e(X_i) = Pr(Z_i = 1|X_i)$$

This is the individual's propensity to be treated (e.g., to smoke) given X_i.

5.2 The Propensity Score Theorem

- **Under Strong Ignorability:**
$$Z_i \perp (Y_i(1), Y_i(0))|e_i$$
- **Under SUTVA and Strong Ignorability:**
$$ATE = E[E(Y_i|e_i, Z_i = 1) - E(Y_i|e_i, Z_i = 0)]$$

STAT 479: Causal Inference – SP26 Lecture Notes

Chenyi Tong

Feb 24th, 2026

1 Big picture: observational studies need assumptions

In observational data, treatment assignment is not randomized, so treated/control groups may differ systematically. Today's focus was the **foundations** we need before doing causal analysis with observational data:

- **SUTVA** (to make potential outcomes well-defined),
- **Ignorability / unconfoundedness** + **Overlap** (to justify "as-if random" comparisons),
- **Matching as design** (to improve covariate balance in practice).

2 Potential outcomes and SUTVA

2.1 Potential outcomes: what would happen under each treatment

For each unit $i = 1, \ldots, N$, define two potential outcomes:

$$Y_i(1), \quad Y_i(0),$$

and let $Z_i \in \{0, 1\}$ indicate treatment status. We only observe one outcome per unit:

$$Y_i = Y_i(Z_i) = Z_i Y_i(1) + (1 - Z_i) Y_i(0).$$

A key estimand is the ATE:

$$\tau = \mathbb{E}\big[Y(1) - Y(0)\big].$$

2.2 Interference and why we need SUTVA

Without restrictions, unit i's outcome could depend on *everyone*'s assignment vector $\mathbf{z} = (z_1, \ldots, z_N)$:

$$Y_i(\mathbf{z}) = Y_i(z_1, \ldots, z_N).$$

Since there are 2^N possible $\mathbf{z}$'s, this would imply 2^N potential outcomes *per unit*, which is not workable for standard identification/estimation.

2.3 SUTVA (Stable Unit Treatment Value Assumption)

SUTVA makes the potential outcomes framework usable in typical settings:

1. **No interference between units:** unit i's outcome depends only on its own treatment,

$$Y_i(z_1, \ldots, z_N) = Y_i(z_i).$$

2. **No hidden versions of treatment:** "$Z = 1$" (and "$Z = 0$") correspond to a single well-defined intervention.

A standard implication is **consistency**:

$$Y_i = Y_i(Z_i).$$

2.4 Latent world vs observed world (identification goal)

It is helpful to distinguish:

- **Latent world (conceptual):** $(V_i, X_i, Y_i(1), Y_i(0))$ where V_i represents unobserved factors.
- **Observed world (data):** (X_i, Z_i, Y_i) with $Y_i = Y_i(Z_i)$.

Goal: **identify** a causal quantity such as $\mathbb{E}[Y(1) - Y(0)]$ using only the observed distribution of (X, Z, Y), which requires assumptions beyond SUTVA.

3 Ignorability (unconfoundedness) and overlap

3.1 Confounding idea

A **confounder** is a variable related to both treatment Z and outcome Y. We collect observed pre-treatment covariates into X, aiming for X to contain all relevant confounders.

3.2 Conditional independence notation (quick reminder)

$$A \perp\!\!\!\perp B \quad \text{means } A \text{ and } B \text{ are independent,}$$

$$A \perp\!\!\!\perp B \mid C \quad \text{means } A \text{ and } B \text{ are independent given } C.$$

Equivalently, $A \perp\!\!\!\perp B \mid C$ implies $\mathbb{P}(A \mid B, C) = \mathbb{P}(A \mid C)$ (when these are well-defined).

3.3 Ignorability / selection on observables

The (strong) ignorability condition for observational studies is:

$$\big(Y(1), Y(0)\big) \perp\!\!\!\perp Z \mid X.$$

Interpretation: *within strata of the same X values*, treatment is assigned "as if" randomly. So two individuals with the same X are comparable in the sense that treatment status does not provide extra information about their potential outcomes.

3.4 Overlap (common support)

We also need treated and control units to exist across the same covariate range:

$$0 < \mathbb{P}(Z = 1 \mid X) < 1.$$

This prevents deterministic treatment assignment within some covariate strata and avoids pure extrapolation.

3.5 Strong ignorability

In lecture we referred to:

$$\textbf{Strong ignorability} \quad \Longleftrightarrow \quad \big(Y(1), Y(0)\big) \perp\!\!\!\perp Z \mid X \ \text{ and } \ 0 < \mathbb{P}(Z = 1 \mid X) < 1.$$

Together with SUTVA, this is the standard assumption set behind many observational estimators.

4 Randomized vs non-randomized: balance and the role of matching

4.1 Balance idea

Randomization implies covariate balance between groups (at least in expectation). For observed covariates X:

$$\mathbb{E}[X \mid Z = 1] = \mathbb{E}[X \mid Z = 0], \quad \text{and often we think of} \quad \mathbb{P}(X = x \mid Z = 1) = \mathbb{P}(X = x \mid Z = 0).$$

In non-randomized settings, these equalities usually fail:

$$\mathbb{E}[X \mid Z = 1] \neq \mathbb{E}[X \mid Z = 0], \quad \mathbb{P}(X = x \mid Z = 1) \neq \mathbb{P}(X = x \mid Z = 0),$$

which signals selection differences in the observed world.

4.2 Matching as a design strategy

Matching tries to construct a comparison sample where covariates are similar across treated and controls, i.e.,

$$\mathbb{E}[X \mid Z = 1] \approx \mathbb{E}[X \mid Z = 0] \quad \text{and} \quad \mathbb{P}(X = x \mid Z = 1) \approx \mathbb{P}(X = x \mid Z = 0)$$

(approximately, for observed covariates).

4.3 Optimal pair matching setup (1:1 matching)

Let treated units be $\{i : Z_i = 1\}$ and controls be $\{j : Z_j = 0\}$. We aim to match each treated unit i to a control $j(i)$ to make X_i close to $X_{j(i)}$.

A simple objective is to minimize total squared distance:

$$\min_{j(\cdot)} \sum_{i:Z_i=1} \|X_i - X_{j(i)}\|^2.$$

4.4 Mahalanobis distance objective

To account for different scales and correlation among covariates, we can use Mahalanobis distance:

$$\min_{j(\cdot)} \sum_{i:Z_i=1} (X_i - X_{j(i)})^\top S^{-1} (X_i - X_{j(i)}),$$

where S is a sample covariance matrix for the covariates (from the available sample).

Conceptual takeaway: matching is primarily a **design step** to improve covariate balance before comparing outcomes.